

SCIENTIFIC INVESTIGATIONS

Real-Time Identification of Upper Airway Occlusion Using Electrical Impedance Tomography

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Study Objectives: Real-time monitoring of upper airway collapse during sleep could be instrumental for studies in biomechanics of obstructive sleep apnea (OSA) and selecting individualized treatment modalities. Although some imaging techniques are used under sedated sleep, none are available during the entire natural sleep process. We hypothesized that electrical impedance tomography (EIT) can be used for noninvasive continuous imaging of the upper airway during natural sleep and quantifying upper airway collapse in terms of its size.

Methods: After determining surface landmarks to attach the electrodes for monitoring the retroglossal airway, EIT was conducted in 10 healthy participants. As a feasibility test of EIT in detecting upper airway collapse, transient airway closure was induced by the swallowing maneuver. These EIT images were confirmed by simultaneous magnetic resonance imaging (MRI) scans. Subsequently, EIT scans were conducted in 7 healthy participants and 10 patients with OSA under nonsedated sleep to determine whether it could identify upper airway narrowing or collapse. Respiratory events were identified by concurrent polysomnography (PSG).

Results: Swallowing-induced airway closure was identified successfully in all 10 participants on simultaneous EIT and MRI scans. Sizes and positions of the upper airway closures in reconstructed EIT images were well correlated with those in magnetic resonance images. Obstructive hypopnea and apnea were detected successfully by EIT in 10 patients with OSA, and no significant changes in EIT data were observed in 7 healthy participants during concurrent EIT and PSG tests. Additionally, conductivity changes in the airway were greater during obstructive apnea than during hypopnea (64.3% versus 26.3%, respectively; $P < .001$) compared with those during baseline respiration.

Conclusions: EIT could be a useful real-time monitoring device for detecting upper airway narrowing or collapse during natural sleep in patients with OSA. Currently, changes in the upper airway size can be estimated with good accuracy, but shape estimation needs future improvements in the EIT image quality.

Keywords: diagnostic imaging, electric impedance, obstructive sleep apnea, structure collapse, tomography

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BRIEF SUMMARY

Current Knowledge/Study Rationale: Information on the obstruction site and pattern of the upper airway during sleep in patients with obstructive sleep apnea can guide selection of obstruction site-specific treatments, such as upper airway surgeries, oral appliances, and hypoglossal nerve stimulation, possibly leading to improved treatment success. No real-time diagnostic technique, however, is available to identify narrowing and collapse of the upper airway during natural sleep, although some imaging techniques can be used under sedated sleep.

Study Impact: This study showed that electrical impedance tomography can quantify upper airway narrowing or collapse during natural sleep. Electrical impedance tomography could be a useful tool to monitor the upper airway narrowing or collapse in patients with obstructive sleep apnea.

INTRODUCTION

Continuous positive airway pressure (CPAP) is currently the treatment of choice, particularly for patients with moderate to severe obstructive sleep apnea (OSA). As many patients do not accept or tolerate CPAP, however, its long-term effectiveness is limited. Although various interventional strategies have been shown to enhance patient acceptance and adherence to CPAP, it remains a major clinical challenge.^{1,2} Some patients prefer surgery, because they do not have to wear cumbersome devices all night if surgery is successful.³ Because surgeries are rarely curative, however, except for the most invasive techniques of maxillary and mandibular advancement, no

surgical modification is recommended as a first-line treatment for OSA.⁴ Recurrent occlusion occurs within the upper airway between the soft palate and epiglottis during sleep in patients with OSA. The anatomical structures responsible for such obstructions are the soft palate, palatine tonsils, base of the tongue, and lateral pharyngeal walls.⁵ The decision regarding where and how to operate is a major challenge due to the absence of diagnostic studies to identify the obstruction site (retropalatal/velopharyngeal versus retroglossal/oropharyngeal spaces) and pattern (anterior-to-posterior [A-P], lateral, or circular collapse) during natural sleep.⁶ Several tests under sedated sleep, including somnofluoroscopy, ultrafast computed tomography (CT), cine-magnetic resonance imaging (MRI),

and drug-induced sleep endoscopy (DISE), have been used for that purpose.^{7–11} Somnofluoroscopy and ultrafast CT emit potentially hazardous ionizing radiation, thereby limiting lengthy and repetitive use.^{7,12} Cine-MRI has a noisy scanning environment and produces claustrophobic effects in the gantry tube; thus, natural sleep is usually unattainable. High cost is also a significant obstacle for routine use of cine-MRI.¹³ DISE is limited in evaluating a multilevel obstruction due to obscured vision below the uppermost occlusion site. Furthermore, mucosal irritation caused by the endoscope sometimes results in failed sedation or premature awakening during the test. Although airway collapsibility is altered according to the depth of sedated sleep, test results may differ from those obtained during natural sleep.¹⁴ Optical coherence tomography using a probe through the nasoesophageal catheter has also been introduced, but its feasibility during natural sleep has not been sufficiently proven and the procedure is laborious.¹⁵ Collectively, no imaging technique is available for continuous monitoring of upper airway patency during natural sleep. Electrical impedance tomography (EIT) is a suitable technique for continuous monitoring due to its excellent temporal resolution of more than 50 frames per second, although there may be limitations in producing clear spatial resolution images. This is the first paper to apply EIT on the upper airway under the hypothesis that EIT can be a continuous upper airway imaging technique during natural sleep. The primary objective of this study was to estimate the upper airway size changes and identify the relating upper airway collapse at the retroglossal level using EIT.

METHODS

The study was approved by the Institutional Review Board of Gyeongsang National University Hospital (IRB file No. 2014-02-010). Written informed consent was obtained from all study participants.

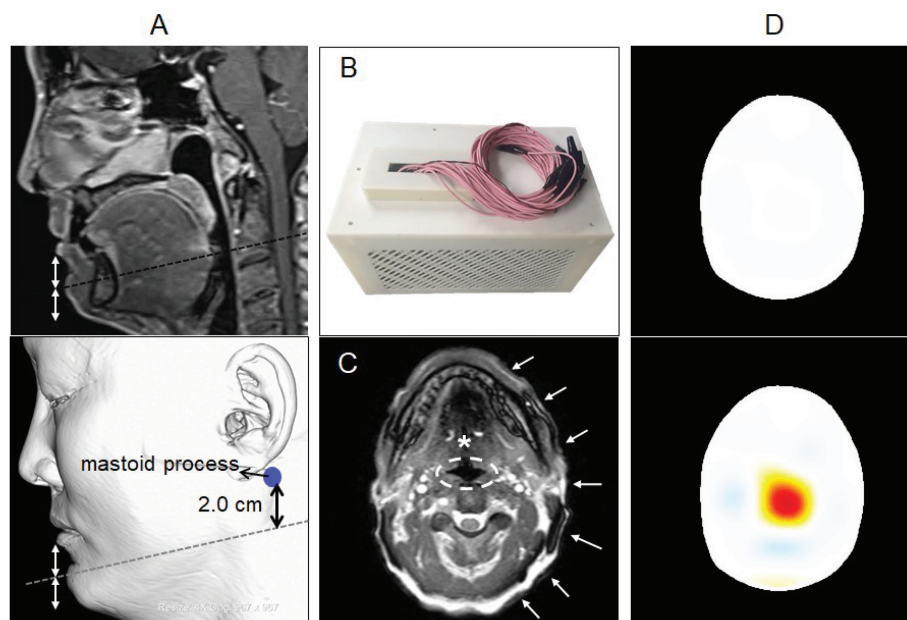
Conductivity and EIT

Electrical conductivity is a material property determining how much current flows by an applied voltage difference. Biological tissues are comprised of cells, extracellular fluid and extracellular matrix materials. Their electrical conductivity is determined by structural properties such as size, shape, density and layout of the cells; composition and amount of extracellular matrix materials and intracellular complexity; and ion concentrations and their mobility. Whereas most biological tissues have their conductivity values in the range of 0.01 to 1 S/m, the conductivity of the air is almost zero because there is a negligible amount of mobile charge carriers in the air.¹⁶ EIT produces cross-sectional images of conductivity distributions inside the human body by injecting insensible amounts of electrical current and measuring induced voltages using multiple surface electrodes.¹⁷ Static EIT imaging cannot produce accurate images inside the body because of the nonlinearity between the conductivity and measured current-voltage data, and low sensitivity of conductivity data inside the human body away from the surface electrodes. However, time-difference EIT imaging, which obtains the voltage difference at different

time points (reference time versus any selected time) shows relative changes in conductivity distribution inside the body in the range of 0 to 100. The value 0 denotes no change compared with the reference value (eg, open airway filled with air). The value 100 denotes a maximum change (eg, completely closed airway filled with soft tissues). By using such relative data, time-difference EIT imaging can provide reconstructed images of the area of remarkable conductivity changes such as transient airway collapse. Because the common errors and noises at two different time points are cancelled out by each other, time-difference EIT imaging could provide reliable data on airway volume changes, as proven with lung ventilation imaging.^{18,19} In this paper, we collect reference voltage data from a participant with his or her upper airway open at a chosen reference time. When the upper airway remains open, the repeatedly measured voltage data from the same participant are equal to the reference voltage data except noise, motion artifacts and conductivity changes subject to blood flow. Thus, time-difference EIT imaging produces no reconstructed airway images during normal respiration. When the upper airway collapses, on the contrary, the measured voltage data from the same participant plummet because the air-filled upper airway (almost zero conductivity value; high impedance) is replaced by soft tissues around the upper airway (high conductivity value; low impedance). The difference between these two data sets is processed to reconstruct upper airway images.

Preliminary EIT Feasibility Study to Detect Airway Occlusion

The surface landmarks for attaching electrodes around the lower face and neck were determined based on a series of mid-sagittal and volume-rendering MRI scans of the head, obtained from dozens of middle-aged men and women. An imaginary line connecting two points (midpoint between the vermilion border of the lower lip and the mentum; the point that is 2 cm below the mastoid process) represented the retroglossal airway space (**Figure 1A**). Ten healthy participants with no history of witnessed apnea (five males, mean age = 21.8 years; range: 20–24 years) were recruited to ascertain the feasibility of EIT for detecting upper airway closure. First, 16 small MRI-compatible carbon-hydrogel electrodes (HUREV Co. Ltd., Wonju, Korea) were attached on the skin of the lower face along the aforementioned line. The landmarks for electrode attachment were validated by simultaneous MRI (Achieva TX; Philips Medical Systems, Best, The Netherlands) and EIT (KHU Mark2.5 EIT system; Kyung Hee University, Seoul, Korea; **Figure 1B**) scans. Each participant was supine and breathing quietly in a neutral position, and then momentarily held his or her breath at the end of a normal inhalation while simultaneous scans were obtained (ie, open airway status). The retroglossal airway space was well visualized on the axial MRI scan when taken along the line of the attached electrodes (**Figure 1C**). Simultaneous MRI and EIT scans were also obtained in the middle of swallowing while holding the tongue at the back of the mouth, which caused transient airway occlusion (ie, closed airway status). Then, the difference in the measured EIT raw datasets between open and closed airway status were processed for EIT image reconstruction (**Figure 1D**). The injected current of 1

Figure 1—Methods and results of our preliminary study.

(A) Representative midsagittal (top panel) and volume-rendering (bottom panel) MRI scans of the head of the same middle-aged man. Dotted lines connecting two points (midpoint between the vermillion border of the lower lip and the mentum; the point that is 2 cm below the mastoid process) corresponded with the retroglossal airway space; thus, they were used as surface landmarks for attaching the electrodes. (B) A photograph of the EIT system used in this study (KHU Mark2.5). (C) A representative axial MRI scan taken while wearing surface MRI-compatible electrodes. The retroglossal space (dotted ellipsoid) behind the tongue (asterisk) and multiple signal voids on the skin along the line of the attached electrodes are visualized in the same plane. (D) A baseline EIT image at the retroglossal space obtained while holding the breath at the end of normal inhalation (top panel) and a representative reconstructed EIT image obtained during swallowing (bottom panel). Increased conductivity in the airway region is visualized as gradated ellipsoids. EIT = electrical impedance tomography, MRI = magnetic resonance imaging.

mA at 50 kHz is known to be safe without causing nerve or muscle stimulation. No significant adverse device effects, such as skin lesions or cardiac arrhythmia, were observed in this preliminary and the following validation studies.

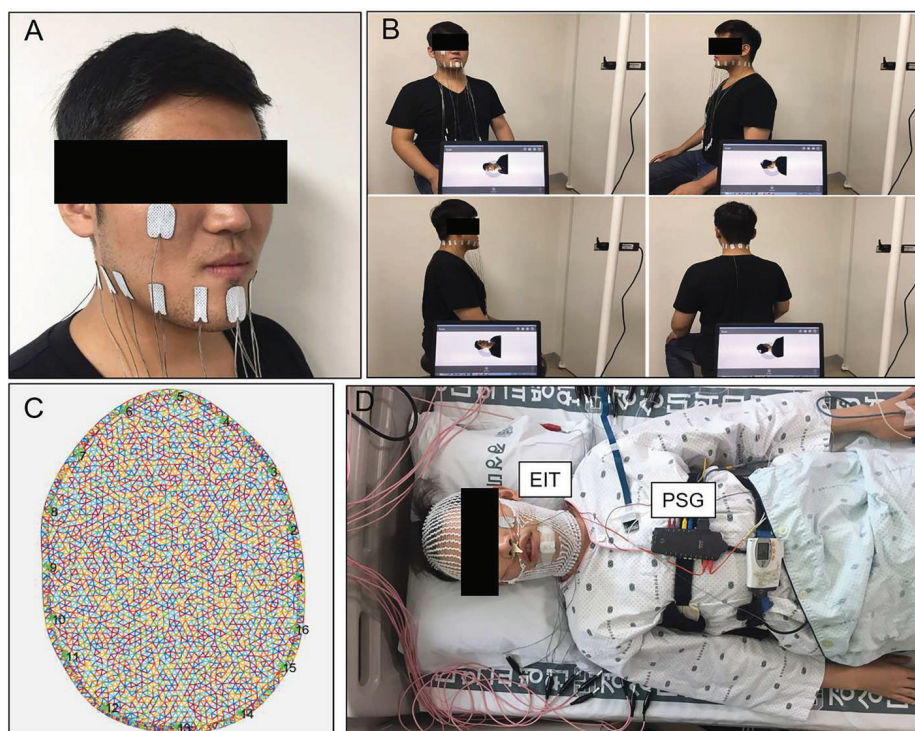
Validation Study to Confirm the Efficacy and Safety of the EIT System

Seven healthy participants (6 males, 1 female) with no history of witnessed apnea, and 10 male patients with OSA were recruited for simultaneous polysomnography (PSG) (Somte, Computimedics, Melbourne, Australia) and EIT (KHU Mark2.5) during sleep. Sixteen small Ag-AgCl electrodes (Blue Sensor BR; Ambu, Ballerup, Denmark) were attached along the lower face according to the aforementioned protocol, with an additional reference electrode attached on the right cheek (**Figure 2A**). Information on the facial boundary shape where the electrodes were attached was obtained using a three-dimensional scanner (Sense; 3D Systems, Inc., Rock Hill, South Carolina, United States) and used for EIT image reconstruction (**Figure 2B** and **Figure 2C**). The participants wore an elastic net bandage (Surginet; Won Industry Co., Gyeonggi-do, Republic of Korea) to avoid displacement of the surface electrodes during sleep (**Figure 2D**). For PSG, airflow signals (using a nasal pressure sensor and an oronasal thermistor), snoring sound, thoracoabdominal respiratory effort signals, electrocardiography, and oxygen saturation were measured. However, electroencephalography (EEG), chin electromyography (EMG), and

electrooculography (EOG) were not measured because those signals were contaminated by externally injected electrical currents from the EIT machine. Therefore, sleep was not staged and arousal was not scored. Respiratory events were scored according to standard clinical criteria (apnea: $\geq 90\%$ reduction in flow; hypopnea: $\geq 30\%$ reduction in flow with $\geq 3\%$ desaturation) except that hypopnea was scored only based on oxygen desaturation, not on arousals. Therefore, apnea-hypopnea index (AHI) in this study was underscored. A hypnotic drug (zolpidem, 10 mg) was used in cases where initiating sleep was difficult. First, a half dose was given to the participant, and an additional half was provided as needed.

Analyses

To remove artifacts, transient (< 1.25 seconds) conductivity changes and/or those with low amplitude ($< 20\%$ of the maximal conductivity change) were filtered. After removing artifacts, conductivity changes within the upper airway region during respiratory events were imaged with reference to the conductivity value in the preceding baseline respiration. Signals from PSG and EIT were synchronized by the trigger signals from the EIT system so that both types of signal could be compared during baseline respiration and respiratory events. The Mann-Whitney *U* test was used to compare age and body mass index (BMI) between groups, and percentage changes in electrical conductivity within the upper airway region between respiratory events, that is, obstructive hypopnea versus apnea,

Figure 2—Validation study procedures.

(A) Attachment of surface electrodes to the lower face. (B) Acquisition of the boundary shape on which the surface electrodes are attached using a three-dimensional scanner. (C) Finite elements method model of surface electrodes being processed based on the obtained three-dimensional images. Extracted boundary information is subsequently used for image reconstruction. (D) A participant wearing both EIT and polysomnography devices in the supine position. An elastic net bandage was applied to the head to avoid displacement of the surface electrodes during sleep. EIT = electrical impedance tomography, PSG = polysomnography.

with reference to baseline respiration. All statistical analyses were conducted using SPSS for Windows software (version 21.0; SPSS Inc., Chicago, Illinois, United States). A value of $P < .05$ was considered significant.

RESULTS

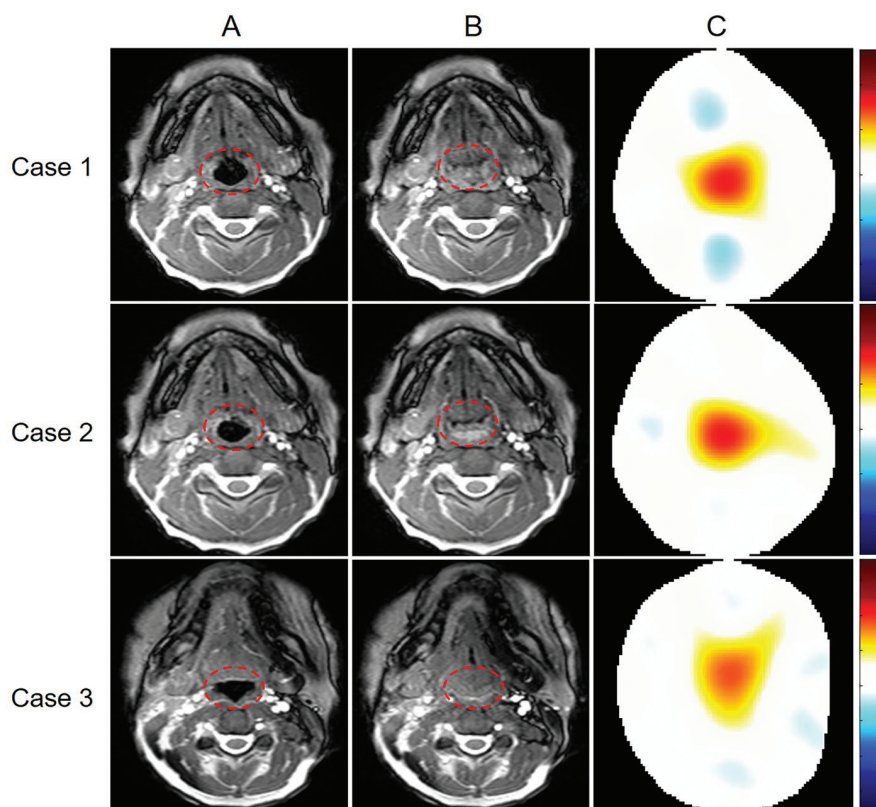
Preliminary Study

The EIT scans showed increased conductivity values in the airway region during the swallowing maneuver (ie, closed airway status) versus the reconstructed conductivity images in open airway status (**Figure 1D**). In all 10 participants, the closed retroglossal space was well identified by the MRI scans obtained during the swallowing maneuver. Although the EIT produced images with a rather blurry boundary, the airway closing shown in the reconstructed EIT image matched well with the corresponding MRI scan obtained at the end of normal inhalation: laterally ellipsoidal shapes of case 1 or 2, and a triangular shape of case 3, respectively (**Figure 3**).

Validation Study

The mean age of the patients with OSA was higher than that of the healthy participants (41.6 versus 25.1 years; $P < .001$). Mean BMI also tended to be higher in the patients with OSA compared with the healthy participants, although the difference

was not significant (26.1 versus 24.2 kg/m², $P = .19$; **Table 1**). Six of the 7 healthy participants, and 7 of the 10 patients with OSA, asked for a hypnotic drug to initiate sleep. Additionally, two (P9, P10) of the three patients with OSA who spontaneously fell asleep woke up in the middle of the test and then took a hypnotic drug because they had difficulty falling back to sleep. However, there were no remarkable changes in the pattern of respiratory events between periods before and after the intake of hypnotic drugs for either patients (**Table 1**). No adverse device effects, including skin discomfort, skin lesions on the surface where the electrode was attached, or cardiac arrhythmia were observed in any of the participants (**Table 1**). No hypopnea or apnea was recorded by PSG in the seven healthy participants. In obtained EIT data, some changes in conductivity were detected, but they were transient or low in amplitude and had no temporal association with respiratory flow changes. Therefore, it was concluded that there exist no significant changes in the reconstructed conductivity values in airway regions of the seven healthy participants (**Table 2** and **Figure S1** in the supplemental material). In contrast, significant increases in the reconstructed conductivity values in upper airway regions were observed during respiratory events in 4 patients with moderate OSA (AHI 15 to < 30 events/h) and 6 patients with severe OSA (AHI ≥ 30 events/h) (**Table 1**). An abrupt increase in reconstructed conductivity value was observed at the beginning of obstructive apnea and this increase in value

Figure 3—Comparison of MRI and EIT scans in three representative healthy participants.

(A) MRI scans obtained while holding the breath at the end of normal inhalation (ie, open airway). (B) MRI scans obtained in the middle of the swallowing maneuver (ie, closed airway). (C) EIT images, reconstructed based on subtraction of the conductivity data on open airway status from that of closed airway status in each participant. Red and blue colors in the right-sided scale bar correspond to increased and reduced conductivity, respectively. The shape of the airway in EIT images (column C) reflected the airway shape shown in MRI scans during open airway (column A). EIT = electrical impedance tomography, MRI = magnetic resonance imaging.

Table 1—Demographics of healthy participants and patients with obstructive sleep apnea.

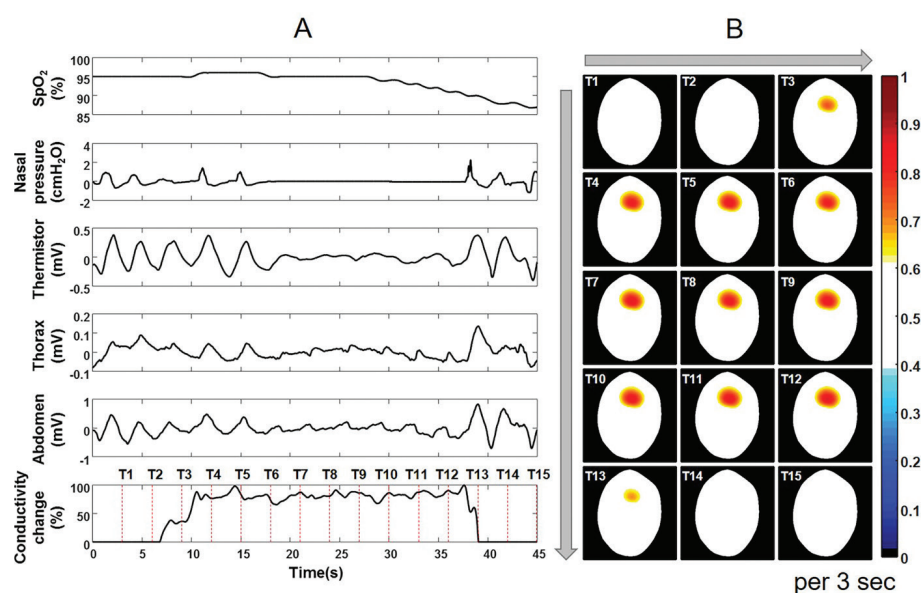
Participant	Sex	Age	Height (cm)	Weight (kg)	BMI	AHI	Zolpidem (mg)	ADE
H1	M	25	174.0	83.8	27.7	0	5	None
H2	M	24	169.0	63.0	22.1	0	5	None
H3	M	26	173.9	65.9	21.8	0	10	None
H4	M	25	173.0	88.0	29.4	0	5	None
H5	M	23	179.0	71.0	22.2	0	0	None
H6	M	28	181.0	76.0	23.2	0	10	None
H7	F	25	169.0	64.9	22.7	0	5	None
Mean (SD)		25.1 (1.6)	174.1 (4.6)	73.2 (9.7)	24.2 (3.1)	0 (0)		
P1	M	39	173.0	80.0	26.7	51.7	5	None
P2	M	56	170.0	74.0	25.6	29.2	10	None
P3	M	36	168.4	72.0	25.4	36	10	None
P4	M	31	185.9	87.9	25.4	23.5	5	None
P5	M	33	183.6	73.2	21.7	25.2	10	None
P6	M	53	171.4	75.6	25.7	52.7	10	None
P7	M	37	171.2	75.7	25.8	57.1	0	None
P8	M	53	159.8	68.0	26.6	19.7	10	None
P9	M	45	163.1	70.6	26.6	68.8	0 → 5	None
P10	M	33	168.3	89.3	31.5	89.3	0 → 5	None
Mean (SD)		41.6 (9.4)	171.5 (8.1)	76.6 (7.1)	26.1 (2.4)	45.3 (22.6)		

ADE = adverse device effect, AHI = apnea-hypopnea index, BMI = body mass index, H = healthy participant, P = patient with obstructive sleep apnea, SD = standard deviation.

Table 2—Percentage changes in electrical conductivity within the upper airway region with reference to baseline respiration in healthy participants and patients with obstructive sleep apnea.

Participant	Conductivity Change	Participant	Conductivity Change		P
			Obstructive Hypopnea	Obstructive Apnea	
H1	0.73 ± 3.58	P1	41.5 ± 17.9	84.6 ± 7.7	
H2	3.97 ± 8.72	P2	25.2 ± 8.9	41.3 ± 17.3	
H3	1.08 ± 2.21	P3	20.8 ± 9.7	76.0 ± 9.0	
H4	0.81 ± 2.65	P4	41.1 ± 21.7	52.1 ± 24.7	
H5	1.35 ± 4.38	P5	8.6 ± 3.1	67.9 ± 9.6	
H6	2.60 ± 5.05	P6	44.6 ± 13.9	53.0 ± 17.0	
H7	6.08 ± 5.26	P7	7.8 ± 9.1	79.1 ± 12.3	
		P8	10.5 ± 6.5	60.5 ± 23.1	
		P9	38.1 ± 20.4	56.8 ± 21.0	
		P10	24.9 ± 14.5	71.9 ± 18.1	
Mean	2.37 ± 4.55	Mean	26.3 ± 14.4	64.3 ± 13.8	< .001*

Values are presented as mean ± standard deviation. * = Mann-Whitney *U* test. H = healthy participant, P = patient with obstructive sleep apnea.

Figure 4—Representative EIT data during obstructive apnea.

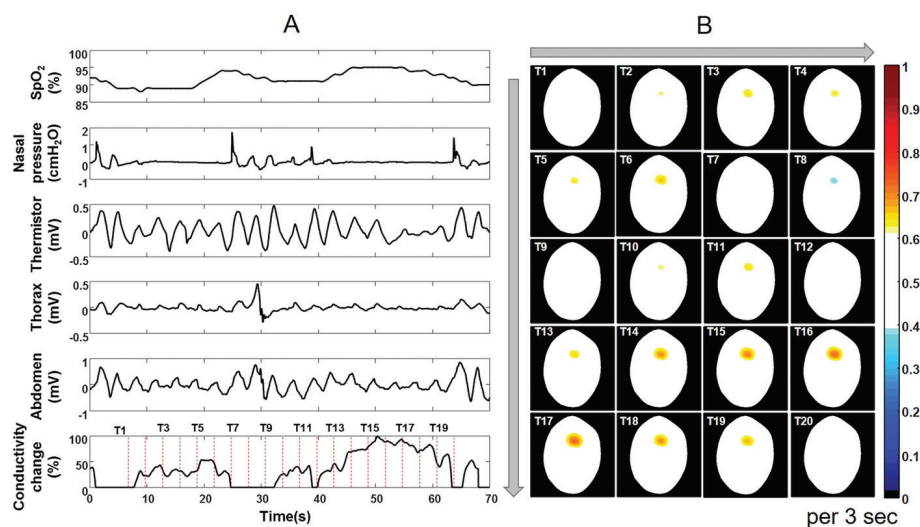
(A) Synchronized polysomnography (pulse oximetry [SpO₂, %], nasal airflow [pressure, cmH₂O], oronasal airflow [thermistor, mV], and thoracic and abdominal effort [each, mV]) and EIT data (conductivity change, %). (B) Sequential changes in reconstructed EIT images reflecting percentage changes in conductivity within the airway region (T1–T15, 3-second time interval). In the scale bar on the right, red and blue colors correspond to increased and reduced conductivity, respectively. Significant increases in electrical conductivity are well visualized during obstructive apnea (T3–T13) compared with those seen during baseline respiration (T1–T2, T14–T15). EIT = electrical impedance tomography.

was sustained during the entire period of ceased airflow in patients with OSA (**Figure 4** and **Video 1** in the supplemental material). Similarly, the reconstructed conductivity value also increased during obstructive hypopnea, but the amplitude of the increase was lower than that during obstructive apnea with reference to the value obtained during baseline respirations (26.3% versus 64.3%, $P < .001$; **Figure 5**, **Table 2**, and **Video 2** in the supplemental material). Interestingly, the conductivity value also increased during central apnea shown in two patients with OSA (P5 and P10; **Figure 6** and **Video 3** in the supplemental material). The mean rate of conductivity change at the beginning of central apnea, however, was slower than that

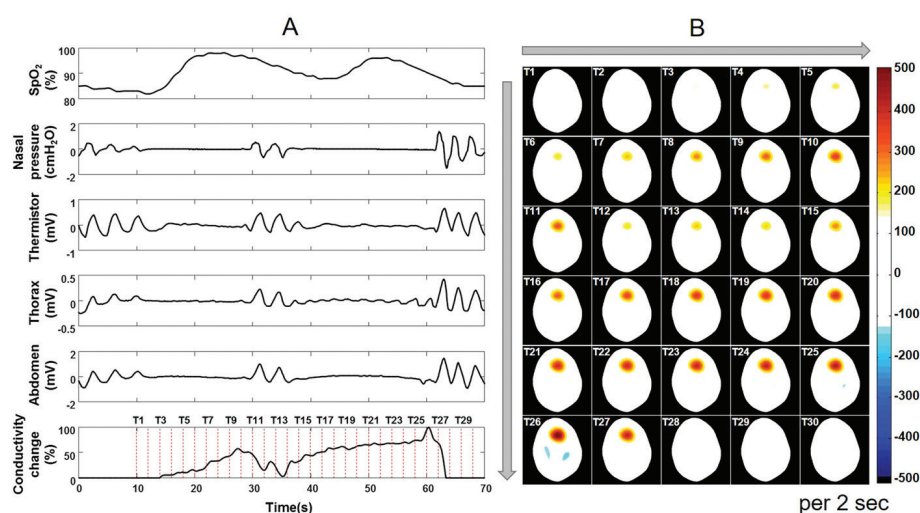
during obstructive hypopnea or apnea (4.3 versus 68.2 %/s and 4.5 versus 9.0 %/s in the P5 and P10 participants, respectively).

DISCUSSION

EIT is a noninvasive, radiation-free imaging technique used to detect changes in the tissue-specific distribution of bioimpedance, determined by the structure and composition of tissues, via electrodes on the body surface.²⁰ Since its first clinical application three decades ago, to produce tomographic images of conductivity changes inside the human body, several clinical

Figure 5—Representative EIT data during obstructive hypopnea and subsequent apnea.

(A) Synchronized polysomnography (pulse oximetry [SpO₂, %], nasal airflow [pressure, cmH₂O], oronasal airflow [thermistor, mV], and thoracic and abdominal effort [each, mV]) and EIT data (conductivity change, %). **(B)** Sequential changes in reconstructed EIT images reflecting percentage changes in conductivity within the airway region (T1–T20, 3-second time interval). In the scale bar on the right, red and blue colors correspond to increased and reduced conductivity, respectively. Compared with those seen during baseline respiration (T1, T7–T9), significant increases in electrical conductivity are visualized during obstructive hypopnea (T1–T7), and to a greater degree during subsequent obstructive apnea (T14–T19). EIT = electrical impedance tomography.

Figure 6—Representative EIT data during central apnea and subsequent obstructive apnea.

(A) Synchronized polysomnography (pulse oximetry [SpO₂, %], nasal airflow [pressure, cmH₂O], oronasal airflow [thermistor, mV], and thoracic and abdominal effort [each, mV]) and EIT data (conductivity change, %). **(B)** Sequential changes in reconstructed EIT images reflecting percentage changes in conductivity within the airway region (T1–T30, 2-second time interval). In the scale bar on the right, red and blue colors correspond to increased and reduced conductivity, respectively. Significant increases in electrical conductivity were visualized during central apnea (T3–T11), although the rate and amplitude of conductivity changes were slower and lower, respectively, than those seen during obstructive apnea (T13–T27). EIT = electrical impedance tomography.

applications, including measurement of pulmonary or gastrointestinal function, diagnosis of breast tumors, and brain imaging, have been attempted.^{21,22} In particular, quantifying regional ventilation using EIT is sufficiently accurate in homogeneously and nonhomogeneously ventilated lungs, and the method has thus gained acceptance as a valuable monitoring tool for critically ill patients under mechanical ventilation.¹⁹ Because the electrical conductivity value of air is very low compared with those of other tissues, as mentioned previously, it is likely that

quantitative measurements of volumetric changes in the air-filled space can be readily obtained by EIT, as shown in the regional lung ventilation monitoring device.^{23,24} Given that mechanical obstruction of the upper airway is a principal phenomenon of OSA, it is postulated that EIT is a suitable technique to monitor upper airway collapse during sleep in patients with OSA.

Identifying the obstruction site and pattern in the upper airway during sleep is an important issue when planning OSA

treatment. Treatment success is believed to be improved by selecting an individualized treatment modality.⁷ In the current study, EIT consistently detected upper airway occlusion induced by the swallowing maneuver (**Figure 1D**). Based on these results, further studies were planned to identify the upper airway closure by EIT during respiratory events in patients with OSA. Although recruited muscles during swallowing differ from those during the respiratory events, similar outcomes were predicted because EIT data are mainly affected by the dimension of the upper airway, not by the surrounding tissues such as muscles. Consequently, the upper airway narrowing and collapse during respiratory events, including obstructive hypopnea and apnea, were successfully detected by EIT under nonsedated sleep in patients with OSA (**Figure 4** and **Figure 5**). The mean values of the percentage change in EIT data were significantly different between obstructive hypopnea and apnea (**Table 2**). There were significant overlaps in EIT data between obstructive hypopnea and apnea, because scoring was not based on anatomical parameters, such as the cross-sectional area of the upper airway, but on measured airflow values. Obstructive apnea does not always indicate complete occlusion of the upper airway, and the cross-sectional area of the airway may change continuously during obstructive apnea, presumably due to alterations in intrapharyngeal negative pressure. To reconstruct the EIT image of the airway collapse, the EIT data obtained during baseline respiration (open airway) was subtracted from that obtained during collapse (closed airway). This seems counterintuitive, as the actual airway cannot be seen when completely closed. Note that, by switching the datasets, the image contrast is simply reversed to reveal airway collapse.

As mentioned earlier, information on the obstruction site and pattern under natural sleep may guide the selection of obstruction site-specific treatments, such as upper airway surgery, oral appliances, and hypoglossal nerve stimulation, possibly leading to improved treatment success in patients with OSA.⁶ For example, hypoglossal nerve stimulation may be the first choice for patients with predominant retroglottal space collapse;²⁵ surgery to enlarge the lateral dimension and increase tension of the lateral pharyngeal wall, such as lateral pharyngoplasty or expansion sphincter pharyngoplasty, might be the first choice for patients with predominant lateral collapse at the retropalatal space alone.^{26,27} Future studies should focus on identifying the morphological changes to provide information regarding the pattern of upper airway obstruction, ie, A-P, lateral, or circular collapse.

Real-time monitoring of the upper airway can also be instrumental in understanding the mechanics of OSA. With a temporal resolution of ≥ 50 frames per second, EIT may provide information regarding the type of obstruction site and preceding pattern in cases of multilevel and multidirectional collapse, or retropalatal versus retroglottal and A-P versus lateral collapse. Because changes in the pharyngeal airway are rapid, detailed information on airway movement is difficult to obtain even by ultrafast CT or cine-MRI. If such information becomes available, it could help us to understand the mechanics of the upper airway in patients with OSA. Further studies should be needed to ascertain whether multi-level EIT imaging can identify the obstruction site and pattern.

A gradual increase in conductivity was observed during central apneic events in the current study (**Figure 6**). This is conceivable because the mechanism of central and obstructive sleep apnea are known to be tightly intertwined. According to previous studies, central apneic events are often associated with upper airway narrowing or occlusion in the absence of negative intraluminal pressure.^{28,29} Moreover, airway occlusion during central apnea was typically gradual, requiring more than 10 seconds for complete occlusion.²⁸ Similarly, gradual increases in conductivity during central apneic events were observed in the current study, whereas abrupt increases were noted during obstructive apneic or hypopneic events. If discrimination of central respiratory events from obstructive ones becomes feasible by using the upper airway EIT, this would help enhance the clinical applicability of out-of-center sleep testing, particularly using peripheral arterial tonometry, which does not directly measure oronasal airflow or respiratory efforts.³⁰

The current study was limited because the EIT measurements were done only at the retroglottal space. A further study of multilevel EIT measurements, at both the retropalatal and retroglottal level, is needed to validate the efficacy of EIT for differentiating obstruction sites, in combination with simultaneous EIT and cine-MRI during sleep. Furthermore, in this study, hypnotic drugs were used in patients having difficulty falling asleep. Given that these drugs can affect upper airway closure, the participants should be monitored under natural sleep in the future study. A motion-related artifact is another critical issue related to the clinical application of EIT.³¹ Body motion significantly affects the voltage values measured by surface electrodes. In the current study, gel-type electrodes stabilized by an elastic net were used to achieve good electrode contacts with the skin, thereby reducing motion artifacts. However, to enhance clinical applicability, other types of electrodes, such as nanoweb fabric electrodes, are desirable.³² In addition, use of a motion sensor may be helpful to alleviate motion-related artifacts by providing information on measured voltage changes induced by body motion and not by airway narrowing. Last, EEG, chin EMG and EOG were not measured during PSG due to signal contamination by externally injected electrical currents from the EIT machine. The EIT machine injects sinusoidal currents at 50 kHz. Because the frequencies of EEG, EMG and EOG signals are much lower than 50 kHz, this interference will be rejected in the future study by adopting advanced filtering methods.

In conclusion, EIT can be used as a noninvasive continuous monitoring technique to identify upper airway narrowing and collapse during natural sleep in patients with OSA. Future studies are needed to correctly identify the site and shape of upper airway collapse.

ABBREVIATIONS

AHI, apnea-hypopnea index
 A-P, anterior-to-posterior
 BMI, body mass index
 CPAP, continuous positive airway pressure
 CT, computed tomography

DISE, drug-induced sleep endoscopy
EIT, electrical impedance tomography
MRI, magnetic resonance imaging
OSA, obstructive sleep apnea
PSG, polysomnography

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